

1. Prove the quadratic equation $ax^2 + bx + c = 0$ have not more than two roots.

2. Determine the nature of roots of $x^2 + 10x + 25 = 0$

3. Find the quadratic equation whose roots are -3 and 5

4. Form and equation, whose roots are thrice the roots of $3x^2 - 8x - 4 = 0$

5. If the roots of equation $5x^2 - kx + 45 = 0$ are equal, find the value of k

6. If the equation $x^2 + 2(k+2)x + 9k = 0$ has equal roots. Find the value of k .

If α and β are the roots of equation $px^2 + qx + r = 0$, prove that

$$\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{q}{p}} = 0$$

7. If one root of the equation $ax^2 + bx + c = 0$ be four times the other. Prove that $4b^2 = 25ac$

8. If α and β be roots of equation $ax^2 + bx + c = 0$, find the value of

$$\alpha^2 + \beta^2$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

1. Evaluate $\sqrt{-36} \times \sqrt{-169}$

9. What are the cube roots of unity? Show that

$$(1 - \omega + \omega^2)^4 + (1 + \omega - \omega^2) = -16$$

10. If ω be the cube roots of unity, prove that

$$(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8) = 9$$

11. Define absolute value of complex number. If $\frac{1-ix}{1+ix} = a - ib$, prove that $a^2 + b^2 = 1$

12. If Z and W are complex numbers whose that $|z| + |w| \geq |z + w|$

13. Express the complex number $-\sqrt{3} + i$ in polar form.

14. Define a complex number, if $\sqrt{x+iy} = a+ib$, prove that $\sqrt{x-iy} = a-ib$

15. State and prove that De-Moivre's Theorem.

16. If $a + ib = \frac{1+i}{1-i}$, prove that $a^2 + b^2 = 1$

17. Apply Demovier's theorem to compute the cube roots of unity

18. State De-movier's theorem. Use it to find the square roots of

$$\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

19. Find the square roots of $(-5-5i)$ using De-movire's theorem.

20. Find the square roots of $4 + 4\sqrt{3}i$ by using De-Moivre's theorem.

21. Find the fourth roots of i

22. Using De-Moivre's theorem. Find the square roots of $2 + 2\sqrt{3}i$

23. If $A = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 3 \\ 3 & 1 \\ 1 & 2 \end{pmatrix}$ find the matrix of $AB - 3I$, where I is a unit matrix of order 2×2

24. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$, show that $A^2 - 2A - 5I = 0$, where I is a unit matrix of order 2×2 and O is null matrix of order 2×2

25. If $A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$, find matrix X such that $A - 3X = \begin{pmatrix} 3 & 5 \\ -8 & 2 \end{pmatrix}$

26. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, verify that $(AB)^T = B^T A^T$

27. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$, verify that $(AB)^T = B^T A^T$

28. If $A = \begin{pmatrix} 1 & 6 \\ 2 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 0 \\ -2 & 7 \end{pmatrix}$, verify that $(AB)^T = B^T A^T$

29. Construct 3×3 matrix whose elements are given by $a_{ij} = 3i - 2j$

30. When will matrix be inverse of each other? show that the matrices $\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix}$ and $\begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix}$ are inverse of each other and verify that their transpose are also inverse of each other.

31. When does a matrix have its inverse? If

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -3 \\ 0 & 3 & -1 \end{bmatrix}, \text{ Find its inverse}$$

32. Find inverse of

$$a) A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 4 \\ 6 & 9 & 7 \end{bmatrix} \quad b) A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 3 & 6 \\ 3 & -2 & 4 \end{bmatrix}$$

33. Prove that

$$a) \begin{bmatrix} a+x & b & c \\ a & b+y & c \\ a & b & c+z \end{bmatrix} = xyz \left(1 + \frac{a}{x} + \frac{b}{y} + \frac{c}{z} \right)$$

$$b) \begin{bmatrix} a & b & ax+by \\ b & c & bx+cy \\ ax+by & bx+cy & 0 \end{bmatrix} = (b^2 + ac)(ax^2 + 2bxy + cy^2)$$

$$c) \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{bmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$d) \begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix} = (x-y)(y-z)(z-x)$$

$$e) \begin{bmatrix} 1 & x & x^3 \\ 1 & y & y^3 \\ 1 & z & z^3 \end{bmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

$$f) \begin{bmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{bmatrix} = abc(a-b)(b-c)(c-a)$$

$$g) \begin{bmatrix} x & x^2 & yz \\ y & y^2 & zx \\ z & z^2 & xy \end{bmatrix} = (x-y)(y-z)(z-x)(xy + yz + zx)$$

$$e) \begin{bmatrix} a+b+2c & c & c \\ a & b+c+2a & a \\ b & b & c+a+2b \end{bmatrix} = 2(a+b+c)^3$$

34. Solve by using Cramer's rule

$$\begin{array}{lll} x - 2y - z = 7 & x + y - z = 3 & x - 2y - 3z = 3 \\ 2x + y + z = 0 & 2y + z = 10 & x + y - 2z = 7 \\ 3x - 5y + 8z = 13 & 5x - y - 2z = 3 & 2x - 3y - 2z = 0 \end{array}$$

$$\begin{array}{lll} 3x + 5z = 14 & x + 2y - 3z = 9 & x + y = 10 \\ 2x + y - 3z = 3 & 2x - y + 2z = -8 & 3x - y = 6 \\ x + y + z = 4 & 3x - y - 4z = 3 & \end{array}$$

$$\begin{array}{l} x + 4y + z = 18 \\ 3x + 3y - 2z = 2 \\ -7y + z = -7 \end{array}$$

35. Find extreme values of $G(x, y) = 10x + 15y$ subject to $x + 2y \leq 20$, $x + y \leq 16$ and $x, y \geq 0$

36. Maximize and minimize $F = 34x + 6y$ subject to $x + y \leq 6$, $x + y \geq 1$, $1 \leq x \leq 3$

37. Maximize and minimize $z = 12x + 3y$ subject to $x + y \geq 12$, $3x + 2y \geq 25$ and $x, y \geq 0$

38. Maximize $z(x, y) = 2x + 5y$ subject to the given system of inequalities $5x + y \geq 10$, $x + y \geq 4$, $x \geq 0$, $y \geq 0$ (use graph paper)

39. Maximize $F = 2x - y$ subject to $x + y \leq 5$, $x + 2y \leq 8$, $x \geq 0$, $y \geq 0$

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40. Maximize and minimize $F = 9x + 40y$ subject to $y - x \geq 1$, $y - x \leq 3$, $2 \leq x \leq 5$.

41. Maximize and minimize $G = 2x + 3y$, subject to $x + y \geq 12$, $3x + 2y \geq 25$, $x \geq 0$ and $y \geq 0$

42. Find the extreme value of the function $F(x, y)$ defined by $F(x, y) = x + y$ on the convex polygonal region given by the inequalities $2x + y \leq 20$, $2x + 3y \leq 24$, $x \geq 0$ and $y \geq 0$.

43. Determine the maximum and minimum value of objective $F = 16x - 2y + 40$ subject to $3x + 4y \leq 27$, $y \leq 4$, $x \leq 7$, $x \geq 0$, $y \geq 0$

44. Minimize the cost = Rs $20x + Rs 30y$ subject to the limitations $3x + 5y \geq 45$, $2x + y \geq 20$ and $x, y \geq 0$.

2. Conic Sections

2.1 Parabola

1. Find the equation of a parabola in standard form.
2. Find the vertex, focus equation of directrix and length of latus rectum of parabola $y^2 - 4y - 4x - 8 = 0$ (2078)
3. Find the equation of parabola whose vertex and (2,3) and focus at (5,3). (2074 / 2076)
4. Find the equation of parabola whose vertex at (-2,3) and directrix is $x = 4$. (2073)
5. Find the equation of the parabola whose vertex at (-1,2) and directrix $x = 4$. (2071)
6. Prove that the line $lx + my + n = 0$ will touch the parabola $y^2 = 4ax$ if $ln = am^2$.
7. Find the equation of tangent and normal at point (4, 6) of the parabola $y^2 = 9x$

2.2 Ellipse and Hyperbola

1. Find the eccentricity, coordinates of the vertices and foci and also the lengths of the major axis, the minor axis and the latus rectum of ellipse $x^2/25 + y^2/16 = 1$
2. Find the eccentricity and the coordinates of the vertices and the foci of the ellipse $\frac{(x-1)^2}{16} + \frac{(y-2)^2}{4} = 1$
3. Show that $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ represents the equation of an ellipse. Find its centre, vertex, focus, eccentricity and equation of directrix of ellipse.
4. Find the equation of ellipse whose foci are $(\pm 2, 0)$ and the length of the latus rectum is 6.
5. Find equation of ellipse whose latus rectum is 3 and eccentricity is $1/\sqrt{2}$
6. Find the centre, vertices, eccentricity and length of latus rectum of ellipse $25x^2 + 4y^2 = 100$

2.3 Hyperbola

7. Find the vertices, foci and eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$
8. Find the vertices, centre, eccentricity, foci, length of transverse axis, conjugate axis and the length of the latus rectum of the hyperbola $9x^2 - 16y^2 - 18x - 64y - 199 = 0$
9. Determine the equation of hyperbola in standard form with focus at (-5, 0) and vertex at (2,0)

3. Geometry

3.1 Co-ordinate in space

1. Define direction cosines and direction ratios of line. Show that $l^2 + m^2 + n^2 = 1$, l , m and n have their usual meaning.
2. Find direction cosine and direction ratios of line joining the points (1, 3, 4) and (2, 5, 6).
3. If a line makes angle $\alpha, \beta, \gamma, \delta$ with the four diagonals of cube, show that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$
4. Define direction cosine. Find the angle between the lines whose direction cosines are given by l_1, m_1, n_1 and l_2, m_2, n_2 .
5. Find the acute angle between whose direction cosines are connected by relation $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$
6. Find the direction cosines l, m, n of two lines which satisfy the equation $l + m + n = 0$ and $2lm - mn + 2nl = 0$
7. Find the direction cosines of two lines which satisfy the relations $2l + 2m - n = 0$ and $lm + mn + nl = 0$. Also find the angle between the two lines.
8. Define direction cosine. Prove that the lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fml + gnl + hlm = 0$ are perpendicular if $f/g + g/b + h/c = 0$
9. Find the projection of the line AB on CD if the coordinates of the points A, B, C and D are (0, 5, 0) (1, 2, 4), (-1, 3, 0) and (3, 5, 6) respectively.
10. Show that the line joining the points (1, 2, 2) and (4,5,6) is perpendicular to the line joining (5, -3, 4) and (7, -1,1).
11. Show that the line joining points (1, 2, 3) and (-1, -2, -3) is perpendicular to the line joining the points (-2, 1, 5) and (3, 3, 2).

3.2 Planes

- Find the plane where the line through $(1, 5, 11)$ and $(-1, -1, -1)$ meets yz -plane.
- Find the length of perpendicular drawn from point $(1, 3, 4)$ to the plane $3x - 2y + 6z = 0$
- Find the ratio in which zx -plane divides the segment joining $(2, 4, 8)$ and $(1, 6, 7)$
- Find the angle between two planes $3x + 2y - 6z = 7$ and $2x + 3y + 2z - 5 = 0$.
- Find equation of plane through the points $(2, -3, 1)$ and perpendicular to line joining the points $(3, 4, -1)$ and $(2, 1, -5)$
- Find the equation of plane through the point $(2, 2, 1)$ and $(3, 1, 2)$ and perpendicular to the plane $x + 2y + 3z = 5$.
- Find the equation of plane through the intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and perpendicular to the plane $4x + 5y - 3z = 8$
- Find the equation of plane through the points $(1, 2, 1)$, $(2, 2, 2)$ and $(0, 1, 0)$.
- Find the equation of the plane containing the lines through the origin with direction cosines proportional to 2, 1, -2 and 5, 2, -3.

Vectors

- Prove by vector method $(1, -2, 3)$, $(2, 3, -4)$ and $(0, -7, 10)$ are collinear
- Prove by vector method $\vec{i} + \vec{j} + 4\vec{k}$, $2\vec{i} + 5\vec{j} - \vec{k}$ and $3\vec{i} + 8\vec{j} - 6\vec{k}$ are collinear
- Define collinear vectors. Prove that three points with the following vectors are collinear. $\vec{i} + 2\vec{j} + 3\vec{k}$, $-2\vec{i} + 3\vec{j} + 4\vec{k}$, $7\vec{i} + \vec{k}$
- Prove by vector method $(2, 6)$, $(1, 2)$ and $(3, 10)$ are collinear
- Show that $2\vec{i} - 3\vec{j} + \vec{k}$, $3\vec{i} - 5\vec{j} + 2\vec{k}$ and $4\vec{i} - 5\vec{j} + \vec{k}$ are co-planar
- Show that $\vec{i} + \vec{j} + \vec{k}$, $2\vec{i} + 3\vec{j} - \vec{k}$ and $-\vec{i} - 2\vec{j} + 2\vec{k}$ are co-planar
- Prove that the vectors $5\vec{a} + 6\vec{b} + 7\vec{c}$, $7\vec{a} - 8\vec{b} + 9\vec{c}$ and $3\vec{a} + 20\vec{b} + 5\vec{c}$ are coplanar.
- Show that $5\vec{a} + 6\vec{b} + 7\vec{c}$, $7\vec{a} - 8\vec{b} + 9\vec{c}$ and $3\vec{a} + 20\vec{b} + 5\vec{c}$ are co-planar
- Show that the points $(1, 3, -1)$, $(1, 1, 0)$, $(2, 5, 4)$ and $(2, 7, 3)$ are coplanar.
- If $\vec{a} = 3\vec{i} + 4\vec{j}$ and $\vec{b} = \vec{i} - \vec{j} + \vec{k}$, show that $\vec{a} \times \vec{b}$ represents vectors which is perpendicular to both $\vec{a}$ and $\vec{b}$
- Find the area of parallelogram determined by vectors $\vec{i} + \vec{j} + \vec{k}$, $-2\vec{i} + 3\vec{j} + \vec{k}$
- Show that the three points whose vectors are $2\vec{i} - \vec{j} + \vec{k}$, $\vec{i} - 3\vec{j} - 5\vec{k}$ and $3\vec{i} - 4\vec{j} - 4\vec{k}$ forms the sides of right angled triangle. Also show that remaining two angles of triangles are $\cos^{-1}\left(\sqrt{\frac{35}{11}}\right)$ and $\cos^{-1}\left(\sqrt{\frac{6}{41}}\right)$
- Using the vector method find the area of triangle formed by the points $A(1, 1, 1)$, $B(1, 2, 3)$ and $C(2, 3, 4)$
- Prove by vector method that $a/\sin A = b/\sin B = c/\sin C$
- In any ΔABC , prove by vector method $a = b \cos C + c \cos B$
- In any ΔABC , prove by vector method that
$$\cos B = \frac{c^2 - a^2 - b^2}{2ca}$$
- Using vector method, prove
 - $\cos(A + B) = \cos A \cos B - \sin A \sin B$
 - $\cos(A - B) = \cos A \cos B + \sin A \sin B$
 - $\sin(A + B) = \sin A \cos B + \cos A \sin B$
 - $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- If $\hat{a}$ and $\hat{b}$ are unit vectors and θ be the angle between them, prove that
$$\frac{1}{2}|\hat{a} - \hat{b}| = \frac{\sin \theta}{2}$$

ABCDEF is a rectangular hexagon. Express the vectors $\vec{AC}$, $\vec{AD}$ and $\vec{AE}$ in terms of $\vec{AB}$ and $\vec{BC}$. Also find the vectors $\vec{BD}$, $\vec{BE}$ and $\vec{BF}$
- State triangle law of vector addition. If ABCDEF is a regular hexagon. Show that
$$\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} = 6\vec{AO}$$
- If ABCDEF is a regular hexagon, show that $\vec{BC} + \vec{BD} + \vec{BE} + \vec{BA} = 6\vec{BO}$, where O is the center of hexagon
- If AC and BD are diagonals of parallelogram ABCD, prove that $\vec{AB} + \vec{DC} = \vec{AC} + \vec{DB}$
- ABCD is a parallelogram, G is the point of intersection of it's diagonals and if O is any point. Show that
$$O\vec{A} + O\vec{B} + O\vec{C} + O\vec{D} = 4O\vec{G}$$
- Show by vector method the angle between two diagonals of cube is $\theta = \cos^{-1}\left(\frac{1}{3}\right)$

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35. If $\vec{a} = (2, -1)$ and $\vec{b} = (-2, -3)$, find $2\vec{a} + 3\vec{b}$, its magnitude and unit vector along $2\vec{a} + 3\vec{b}$
36. Find unit vector perpendicular to each vectors $\vec{i} + 3\vec{j} - 4\vec{k}$ and $2\vec{i} + \vec{j} - \vec{k}$
37. Using vector method find the area of triangle formed by the points A(1, 1, 1) and B(1, 2, 3) and C(2, 3, 1).

Statistics

- Find the first and third quartile from 2, 5, 7, 10, 20, 16.
- Find the standard deviation for data 12, 14, 16, 18, 20.
- Find the first four moment of the following data 2, 4, 5, 8, 10.
- Calculate the G.M and H.M from the following data

Class	10-20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	3	3	7	4	5

- Find the quartile deviation and its coefficient from the following data

Class	0-20	20 - 40	40 - 60	60 - 80	80 - 100
Frequency	4	8	12	3	5

Class	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	2	4	6	7	3	1	5

- By Using product moment formula. Calculate from the following data

Class	25	19	28	26	20	18	24	20	22	18
Frequency	60	54	66	70	53	?	62	51	65	50

Where average sales is 59 units?

- Find standard deviation and coefficient of variation (CV) of the data

Class	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Frequency	4	6	8	10	4	3

- Find standard deviation and coefficient of variation (CV) of the data

x	5	10	15	20	25
y	2	4	1	6	9

- Find arithmetic mean and standard deviation from the following data

Class	0 - 4	4 - 8	8 - 12	12 - 16	16 - 20	20 - 24
Frequency	7	7	10	15	7	6

- Calculate the Karl Pearson's coefficient of correlation

x	7	3	11	5	9
y	10	12	7	8	6

- Find correlation coefficient between two variables

X	25	19	28	30	18	24
Y	58	52	65	70	51	62

- Find the regression equation of x on y from the following data

x	2	4	5	6	8	11
y	18	12	10	8	7	5

13. Find the regression equation of y on x from the following data

x	2	4	6	8	10	12
y	5	6	13	16	13	24

Also estimate the value of Y when $X = 5$

14. If mean and variance of a binomial distribution are 40 and 36, find the value of p , q and n

15. A bag contains 5 white, 7 red and 8 black balls. If two balls are drawn one by one without replacement, what is the probability that both are white.

16. A bag contains 5 green and 7 red balls. Two balls are drawn at random. What is the probability that one is green and the other is red?

17. 10. Suppose 3 people are selected at random from a group of 7 men and 6 women. What is the probability that 2 men and 1 women are selected.

18. A problem of mathematics is give to the three students, A , B and C and the chances of solving it are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$. Find the probability that problem will solve.

19. There are 10 electric bulbs in the stock of a shop out of which 4 are defectives. In how many ways can a selection of 6 bulbs be made so that 4 of them may be good bulbs.

20. A class consists of 60 boys and 40 girls. If two students are chosen at random what is the probability that

21. Both are boys ii) One boy and one girl

22. A binomial distribution consists of 5 independent trials. If the probabilities of 1 and 2 successes are respectively $\frac{1}{4}$ and $\frac{1}{3}$. Find the probability of success and failure in trial. Also find $P(r = 3)$

23. A coin is tossed 5 times. Find the probability of getting

i) Exactly 2 head ii) no head

24. A coin tossed three times successively. Determine the probability of getting i) exactly 2 heads ii) at least 2 heads and iii) at most 2 heads.

25. The probability that student pass mathematics test is $\frac{3}{5}$ and the passes both math and chemistry test is $\frac{1}{5}$. The probability that he passes at least one test is $\frac{19}{20}$. What is the probability that he passes the chemistry test?

26. Define sampling. A population consists of five number 2, 5, 6, 8, 9. Taking samples size of two, show that the mean of the sampling distribution of sample mean is equal to the population mean.

S.N	Units	Short Question	Long Question	Total Marks
1.	Algebra	$4 \times 2 = 8$	$4 \times 4 = 16$	24
2.	Conic Section	$2 \times 2 = 4$	$2 \times 4 = 8$	12
3.	Geometry	$2 \times 2 = 4$	$2 \times 4 = 8$	12
4.	Vectors	$3 \times 2 = 6$	$2 \times 2 = 8$	14
5.	Statistics and Probability	$3 \times 2 = 6$	$3 \times 4 = 12$	18
Total		$14 \times 2 = 28$	$13 \times 4 = 52$	80

Pass Marks: 32 | Full Marks 80

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